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THE ROLE OF OVERSTOCKING AND POOR STOCK ROTATION IN FOOD WASTE IN SAUDI SUPERMARKETS

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GSEM 201 – Research & Innovation

April 29, 2026

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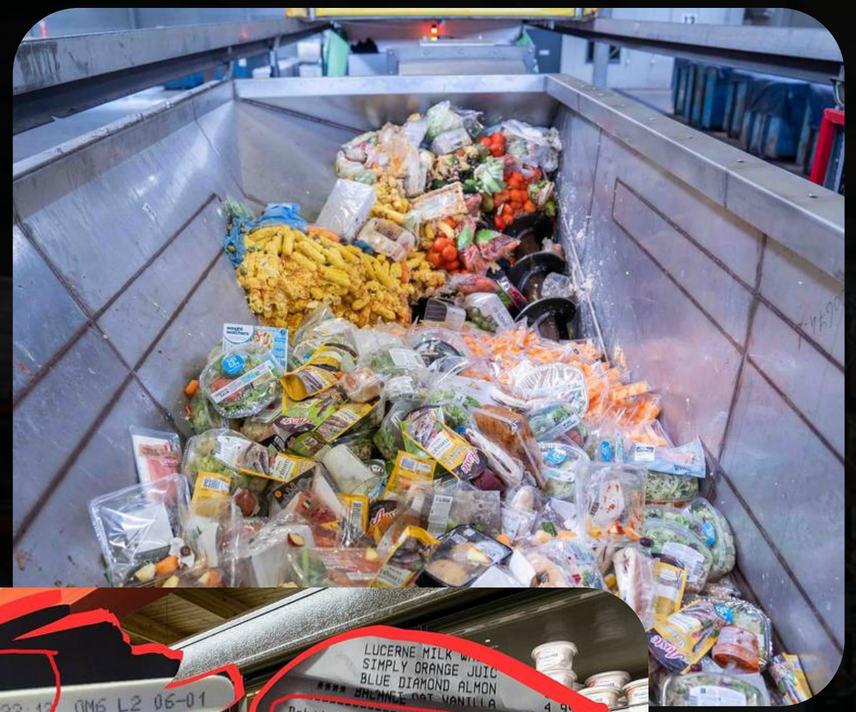
Research Purpose / Question

Research Question:
To what extent do overstocking and poor stock rotation practices contribute to food waste in Saudi supermarkets?

Purpose:

- Examine how supermarket operations—not just consumers—cause food waste
- Assess the link between inventory management and waste generation
- Identify which factor (overstocking or poor rotation) has a greater effect







Background and Literature Review

Key Insights from Literature:

- Around one-third of global food is wasted (Gustavsson et al., 2011)
- Retail waste often stems from inventory management errors (Nikoličić et al., 2021)
- In Saudi Arabia:
  - High import dependency amplifies cost of waste
  - Most studies focus on consumer behavior, not retail operations
  - Few studies measure overstocking and FIFO compliance empirically

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Methodology

Research Design:

- Quantitative cross-sectional survey
- 13 participants from Saudi supermarkets
- Roles: managers, supervisors, and stock staff
- Data collection: online questionnaire (Likert scale)
- Analysis: descriptive statistics + correlations



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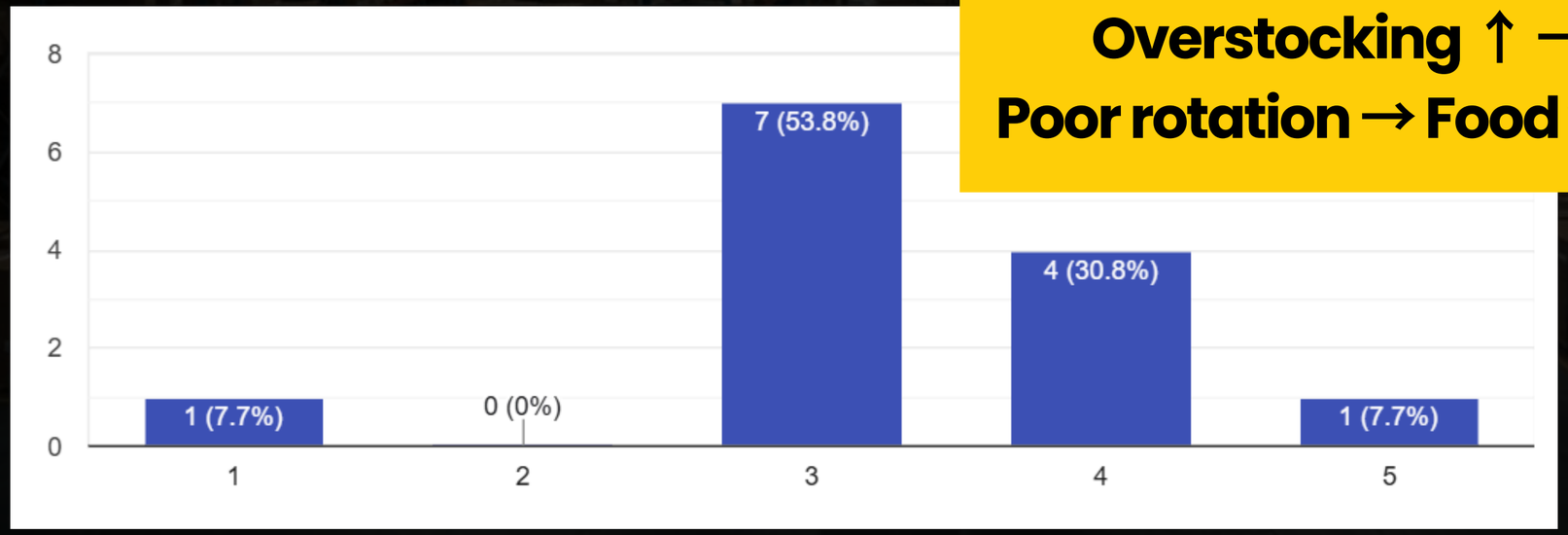
Main Findings

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Findings Overview:

- Overstocking was common in many departments
- FIFO (First-In, First-Out) was practiced—but inconsistently
- Food waste considered a moderate but ongoing issue
- Overstocking showed stronger correlation with waste than poor rotation
- Other issues: poor storage, communication gaps, and unclear expiry dates

Overstocking ↑ → Food waste ↑
Poor rotation → Food waste (less direct)



Food waste is a common issue in my store.

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Discussion / Implications

Implications for Supermarkets:

- Improve demand forecasting and data use
- Strengthen training on stock rotation
- Enhance communication between departments
- Use technology (AI, expiry alerts, forecasting tools) to control waste

Academic Contribution:

- Fills a research gap in Saudi retail food operations
- Highlights store-level decision making as a driver of sustainability

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Conclusion

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Key Takeaways:

- Both overstocking and poor stock rotation lead to food waste
- Overstocking has the greater effect
- Waste results from management habits and communication issues, not just consumer behavior

Future Research:

- Larger samples and multiple supermarket chains
- Use of real inventory and waste data
- Explore impact of smart systems and AI forecasting

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References (APA style)

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THANK YOU SO MUCH

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